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Investigation on the occurrence of mutual interference between pulsed terrestrial LIDAR scanners

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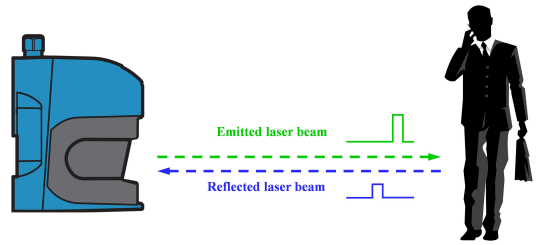
***Abstract*— LIDAR scanners are essential components of intelligent vehicles capable of autonomous travel. Mutual interference between LIDAR scanners has not been regarded as a problem yet. Mutual interference was identified as a problem of increased importance because of the appearance of safety functions and the increasing rate of vehicles equipped with LIDAR scanner. This paper will show the probability that any LIDAR scanner is probably interfered mutually by considering spatial and temporal overlaps. It will present three types of experiments and their results are showed, according to the arrangement of two LIDAR scanners.**

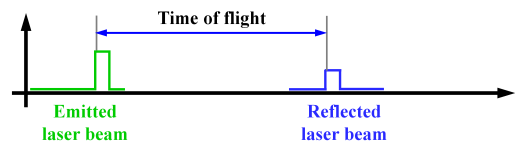
I. Introduction

LIDAR (LIght Detection and Ranging) scanners have been used for many years for obstacle detection. LIDAR scanners operate by sweeping a laser across a scene and at each angle, measuring the range and returned intensity. The obstacle detection system uses a high-performance LIDAR scanner which provides fast single-line LIDAR scans. Histogram analysis of the returned range is used to select obstacle candidates. After candidates are matched and then they are merged with candidates from previous scans. Finally, the positions of all obstacles are updated based on vehicle motion before the next laser scan is acquired. LIDAR scanners are a range scanning system that resolves distance based on the known speed of light ($300\,000\,\mathrm{km\,s^{-1}}$), measuring the ToF (Time-of-Flight) of a laser pulse between the subject for each point. LIDAR scanner emits a laser pulse to an object and wait a reflected laser pulse. The incoming laser pulse experiences a delay depending upon the distance. The distance to the object is calculated by the transit time of a short laser pulsed beam to be reflected and received by the LIDAR scanner as illustrated in Fig. 1. Radar uses this principle of ToF in a similar manner.

Up to now, interference has not been considered as a problem because the percentage of vehicles equipped with LIDAR scanner and therefore the probability of interference was extremely rare, and the scanners were used mainly for comfort functions. On the contrary, obstacle detection functions of autonomous vehicle require very low failure rates. Google's robotic cars have about $150\,000 in equipment including a $70 000 LIDAR scanner [1]. The LIDAR scanner mounted on the top is a Velodyne 64-beam laser. This laser allows the vehicle to generate a detailed 3D map of its environment. The car then takes these created maps and combines them with high-resolution maps of the world, producing different types of data models that allow it to drive itself. So despite a predicted higher number of LIDAR scanners, the probability of interference-induced problems has to be reduced considerably. With the increasing autonomous vehicle equipped with LIDAR scanner for obstacle detection and avoidance to navigate safely through environments, the probability of mutual interference becomes an issue. Imagine such a massive traffic jams in Fig. 2. What if all these vehicles are equipped with LIDAR scanner. With the growing number of autonomous vehicle equipped with LIDAR scanner operated close to each other at the same time, LIDAR scanner may receive signals from other LIDAR scanners. The reception of foreign signals can lead to problems such as ghost targets or a reduced signal-to-noise ratio. The reception of unwanted signals from other LIDAR scanners is called mutual interference between LIDAR scanners.



(a) Normal ToF situation



(b) Normal ToF measurement

Fig. 1: Normal ToF

II. Related work

Within a few years, the penetration rate of vehicular radar systems will have drastically increased in this new emerging market, and for safety-related applications especially the interference risk will threaten further proliferation if harmful mutual interference pops up. Trying to find efficient and pragmatic countermeasures to avoid the apparent interference risk at that point where severe interference problems creating malfunction or out-of-order situation of the safety radar devices have become reality, is too late. The only reasonable

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Fig. 2: A massive traffic jams

and valid approach is to counteract in advance before the problem becomes manifest. The European funding project MOSARIM (MOre Safety for All by Radar Interference Mitigation) started in January 2010 with the objective to investigate possible automotive radar interference mechanisms by both simulation and real-world road-tests. Appropriate countermeasure and mitigation techniques were studied and assessed, and general guidelines and recommendations developed [2]–[5]. The ideas for countermeasures were extracted and structured into six basic categories. From the six basic categories, a list of 22 variants was compiled, evaluated and ranked with respect to various criteria like the implementation effort, required power and computational resources, cost, requirement for harmonization, etc. The top 9 countermeasures were selected for further evaluation, in order to derive guidelines. Their mitigation performance ranges from a few dB up to (theoretically) infinite dB Their accurate and consistent application to all new automotive radar products will result in automotive radar operations that will be close to interference free.

Time-of-flight range cameras acquire a three-dimensional image of a scene simultaneously for all pixels from a single viewing location. It shows how the mixed pixel and multipath separation algorithm can be applied to range imaging measurements projected into Cartesian coordinates. Significant improvements in accuracies can be obtained in particular situations where the influence multipath interference frequently causes distortions [6]–[8]. The multiple return separation algorithm requires a consistent and known relationship between both amplitude and phase response of the cameras at both modulation frequencies. This type of calibration was not available for these cameras, so it was performed by comparing a manually selected regions of the image judged to be minimally affected by multipath interference. If these regions did contain multipath, then the calibration would be compromised leading to poor performance. Another source of potential error is distance linearity measurement inaccuracies common in not calibrated range cameras. It is also possible that the optimization algorithm encountered local minima, producing an incorrect answer.

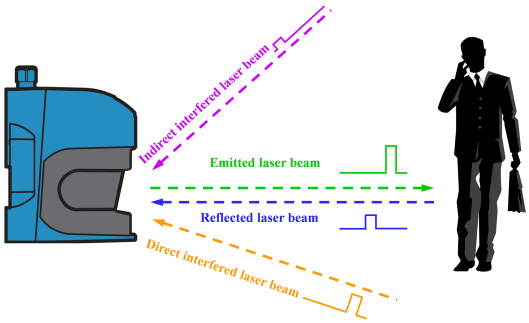
Optical interference between a sensor and another sensor is an essential matter. It is very difficult to separate the amplitude modulated wave from the sensor itself and the one from another sensor after they mixed once. The simplest method for prevention of such interference is that each sensor is installed with tilting them forward 2 to 3 degrees, however that is not enough. Another measure is to emit laser as few as possible. Laser should be off while sensor data is unnecessary. Laser is automatically off at unnecessary area and curtailed scan in accordance with the analyzed result. These functions contribute to not only prevention of interference but also an extension of laser device and reduction of power consumption. Moreover, one more step is prepared, which is a function that motor speed is slightly changed by command. Difference of motor speed that is more than 300 prevent almost the interference [9].

III. Interference of RADAR and LIDAR scanner

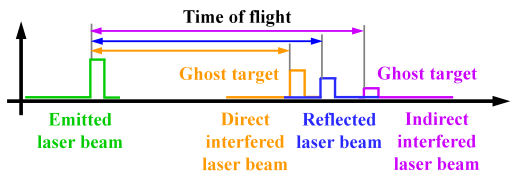
With an increasing number of automotive radar sensors, they operate close to each other at the same time and may receive signals from other radar sensors. The reception of foreign signals (interference) can lead to problems such as ghost targets or a reduced signal-to-noise ratio. These ghosts are particularly bothersome since they move and behave like the real targets (which they echo), and so the receiver has difficulty in isolating the correct target echo. These problems can be overcome by incorporating a ground map of the radar's surroundings and eliminating all echoes that appear to originate below ground or above a certain height.

Interference in general describes the coherent superposition of two or more different waves at one point in space. A wave itself can be defined as a rotating vector in the complex plane, where the addition of several vectors result in deterministic changes of magnitude and phase. The signal amplitude can be increased (constructive interference) or decreased (destructive interference). Concretely, interference effects in automotive radar devices are caused by the superposition of disturbances from other radio devices (e.g. vehicular radar, surveillance radar, fixed links) at the receiving antenna with the incoming coherent use signals being reflected from objects within the radar's detection zone. These induced effects from other radar units in the near vicinity of the victim radar receiver are called mutual interference of radar.

The interference is inherent to all wireless applications sharing the same or an overlapping frequency range. Also automotive radars are exposed to various emissions from other frequency users such as automotive radars in others cars at near distance, other radars roadside or near roads such as speed radars, radars for surveillance purposes e.g. such as surveillance of railroad crossings or surveillance of buildings. Multiple radar systems operate in proximity to each other and thus generate mutual interference. If two or more radars that are in proximity to each other, transmit pulses in a same operating frequency band, each radar system will receive signals directly from the other radar(s) that are more-or-less indistinguishable from reflections from radar targets. These directly-received pulsed interference signals can be of large-amplitude and can tend to saturate receiving systems and target displays. This well-known problem manifests itself in increases in false alarm rates and in undesirable losses of



(a) Interfered ToF situation



(b) Interfered ToF measurement

Fig. 3: Interfered ToF

sensitivity for detecting radar targets. Mutual interference is a serious problem for many types of radar and is expected to be a continuing problem for future radar concepts. Direct interference is occurred from one radar sensor to another but indirect interference by radar reflection on other objects. Fig. 3 shows some scenario of multipath interference. This problem of radar is occurred at the LIDAR scanner in a similar manner.

Interference from an identical pulsed radar sensor with the same PRF (Pulse Repetition Frequency) generates a ghost target at a constant distance. If the radars are similar, except for slightly different PRFs, then a ghost target will appear as a large target moving slowly in the range. The apparent velocity is dependent on the difference between the two PRFs and could be anything from a couple of millimeters per second up to a couple of hundred meters per second. A slightly different PRF results in a moving ghost target since the time difference between the transmitted pulse and the interfering pulse increases or decreases from pulse repetition period to pulse repetition period. If the PRFs are vastly different, then more than one ghost target could appear per cycle [2], [4].

While LIDAR scanner and radar both use similar technologies and approaches to tracking position and movement of objects, there are differences in how each technology works and the types of applications for which each can best be used. Both technologies use energy reflected from objects to determine various aspects of those objects, but the types of energy used in each one is different. The types of objects that can be accurately located and measured by LIDAR scanner and radar are also different in size and nature. Radar units transmit radio waves at a designated frequency that reflect off of a moving target vehicle and return to the unit. The radio waves shoot out in a cone-shaped pattern that covers roughly $75\,\mathrm{m}$ at a range of $300\,\mathrm{m}$. A laser differs from other sources of light because it emits light coherently. Spatial coherence allows a laser to be focused to a tight spot and to stay narrow over long distances. On the other hand, LIDAR scanner sends out a laser beam. The laser beam is very narrow that covers about $1\,\mathrm{m}$ wide at $300\,\mathrm{m}$.

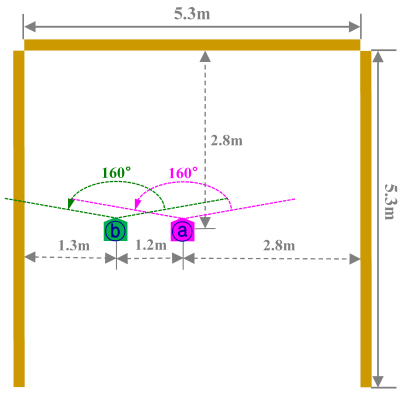
It appears that the LIDAR scanners are very directional so that at any given instance only a small percentage of the angular space around any given units is required to be free to some degree of interference. While the individual laser pulses of the LIDAR scanner are not explicitly coded, they are sensitive to some quality of the reflected light and not just its brightness. The LIDAR scanner cannot uniquely identify the source of the transmission. They would use a broad spectrum of pulse timings so that significantly increased LIDAR scanner illumination merely raises the noise floor slightly rather than risking synchronized sweeps by two nearby vehicles for parts of their field of view. It is probably the case that it is impossible to pack LIDAR scanner equipped cars densely enough that their LIDAR scanners would interfere.

IV. Experiments

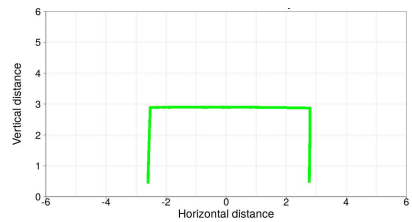
In this section, four types of experiments and their results are showed, according to the arrangement of two LIDAR scanners. Two types are Side by side arrangements, one is front to front arrangement, and the other is back to back arrangement. In each experiment, training the environment carried out first, and then mutual interference testing is done later. The mutual interference between LIDAR scanners was tested on two SICK LMS-511 LIDAR scanners currently available off-the-shelf.

Fig. 4(a), Fig. 5(a), Fig. 6(a), and Fig. 7(a) show the experimental environment. The experimental area is surrounded by low glossy wood walls. Two LIDAR scanners are located inside of the experimental area. One LIDAR scanner - *LIDAR scanner* a - is acting like the victim located at the center and the other LIDAR scanner - *LIDAR scanner* b - acting like the interferer located between LIDAR scanner a and wall. LIDAR scanner b operates only when it is necessary to measure the damage due to mutual interference. LIDAR scanner a runs and records measured data continuously. The analysis of recorded data will show the occurrence and influence of mutual interference.

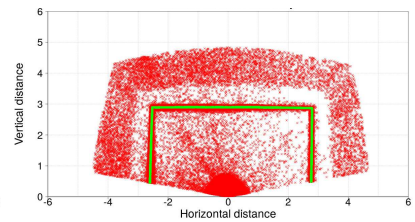
We have conducted four experiments to investigate the occurrence of the mutual interference between LIDAR scanners. In each experiment, the first trial is investigating normal LIDAR scanner operation in the experimental environment. LIDAR scanner a is normal operating state, but LIDAR scanner b is an aborted state. We record the measured distance data at the LIDAR scanner a for 4 hours. Fig. 4(b), Fig. 5(b), Fig. 6(b), and Fig. 7(b) show the distance map acquired by LIDAR scanner a. Green color filled square symbols (■) indicate right distance data. Scatter plots are composed of many superimposed green color filled square symbols show the position of the wall that the LIDAR scanner detect, and they are exactly matched



(a) Experimental setup

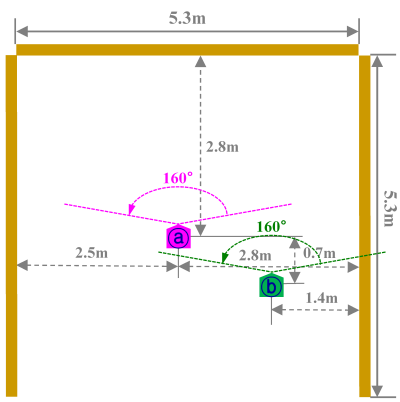


(b) Normal distance map

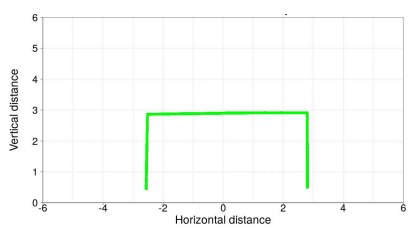


(c) Interfered distance map

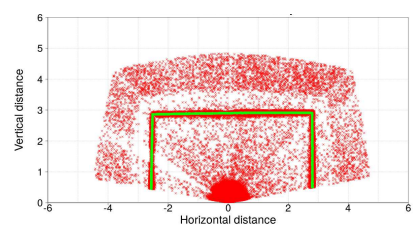
Fig. 4: Side by side arrangement #1



(a) Experimental setup

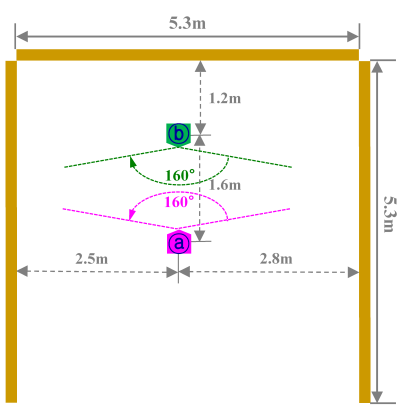


(b) Normal distance map

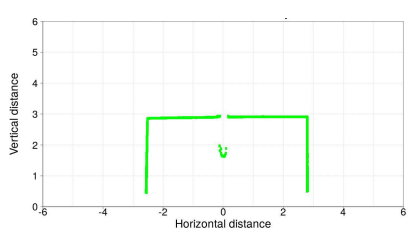


(c) Interfered distance map

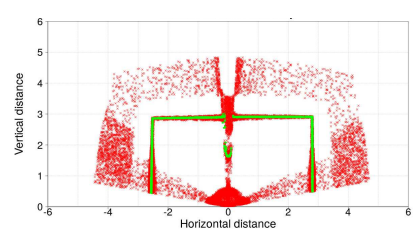
Fig. 5: Side by side arrangement #2



(a) Experimental setup

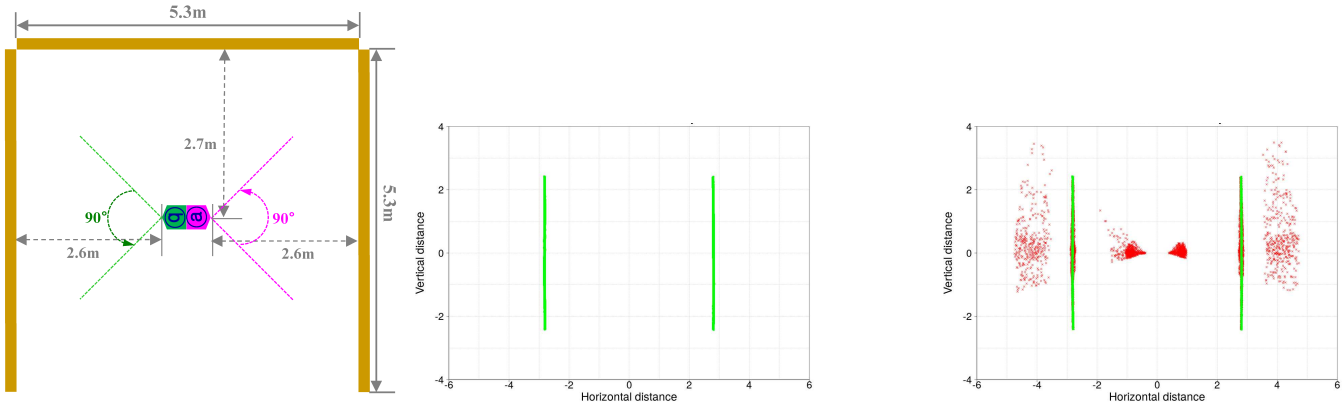


(b) Normal distance map



(c) Interfered distance map

Fig. 6: Front to front arrangement



(a) Experimental setup (b) Normal distance map (c) Interfered distance map

Fig. 7: Back to back arrangement

with the experimental environment in Fig. 4(a), Fig. 5(a), Fig. 6(a), and Fig. 7(a) respectively.

In each experiment, the second trial is investigating abnormal LIDAR scanner operation in the experimental environment. Both LIDAR scanner a and LIDAR scanner b are normal operating condition. We record the measured distance data from the LIDAR scanner a for 24 hours. For 24 hours, total scan lines are 2 160 000, and total scan angles are 1 384 560 000. Fig. 4(c), Fig. 5(c), Fig. 6(c), and Fig. 7(c) show the distance map acquired by LIDAR scanner a. Scatter plots show the mixture of the correct and wrong position of the wall. Green color filled square symbols (■) indicate proper distance data and red color filled cross symbols (X) indicate incorrect distance data. The correct positions are composed of many superimposed green color filled square symbols indicate LIDAR scanner detect correctly. However, the wrong positions are composed of many superimposed red color filled square symbols indicate that one LIDAR scanner affected by the other LIDAR scanner called ghost target.

Tab. II and Fig. 8 show how long the event has occurred the consecutive interfered angles from the result of the second trial in the three experiments. The mutual interference occurs mostly single angle on the same scan line. In some case, the mutual interference is occurred up to 17 consecutive angles. It states that the interference has temporal locality. If a particular angle is interfered at a given time, then it is likely that nearby angles in a line are probably interfered in the near future. Single mutual interference can be ignored as noise or error. It is hard to ignore consecutive mutual interference on the same line because it is possible the real object.

Tab. I and Fig. 8 show how long the event has occurred in the same angles in the consecutive interfered lines from the result of the second experiment. The mutual interference occurs mostly single line on the same scan angle. In some case, the mutual interference is occurred up to 4 consecutive lines at the same scan angle. It states that the interference has spatial locality. If a particular angle is interfered at a given time, then it is likely that the same angle is probably interfered in the near future line. Single mutual interference can be ignored as noise or error. It is hard to ignore consecutive mutual interference at the same angle because it is possible the real object.

TABLE II: Occurrence of consecutive mutual interferences at the same angle

Arrangement	1	2	3	4
Side by side #1	42 194	49	0	0
Side by side #2	230 577	168	1	0
Face to face	35 654	103	1	1
Back to back a	4686	4	0	0
Back to back b	4864	2	0	0

These results state that the interference has temporal and spatial locality. If a particular angle is interfered at a given time, then it is likely that nearby angles in a line are probably interfered in the near future. If a particular angle is interfered at a given time, then it is likely that the same angle is probably interfered in the near future line. It is hard to ignore consecutive mutual interference on the same line or the same angle because it is possible the real object not noise or error.

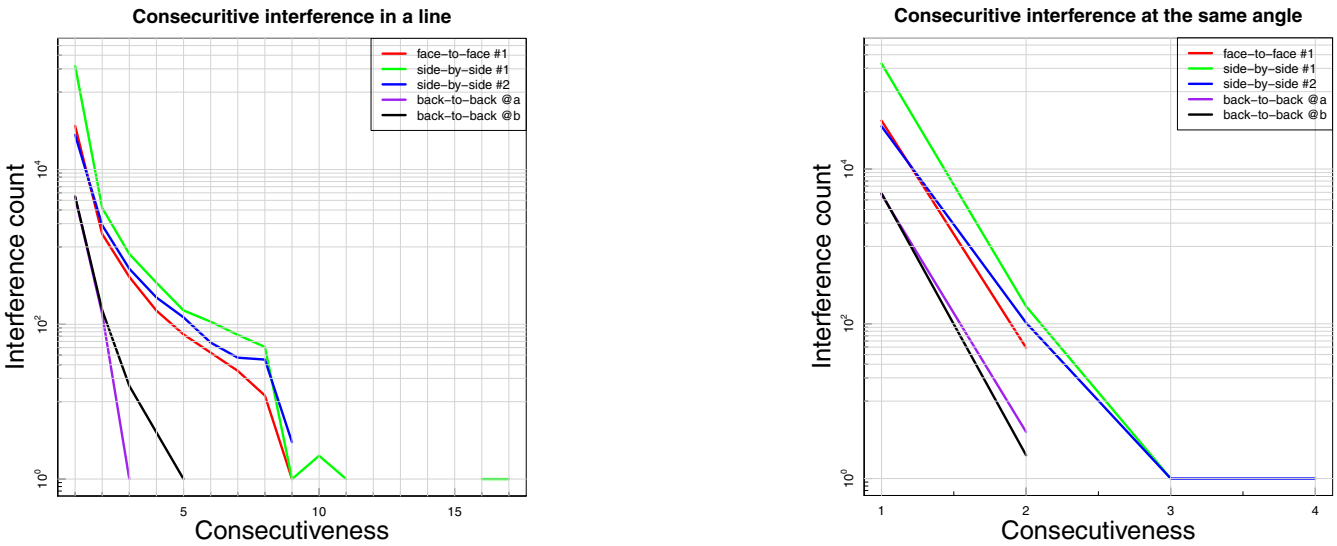
V. Conclusion

Here experimental results regarding the mutual interference between the LIDAR scanners has been presented. Three experiments illustrating the matter are reported. In each experiments, one trial gives a normal distance map of LIDAR scanner. The other trial demonstrates interfered distance map that LIDAR scanner is affected by another LIDAR scanner. Mutual interference of the LIDAR scanner decrease the reliability of the measured data from the LIDAR scanner. It may make serious faults because the obstacle detection functions of autonomous vehicle rely on heavily the LIDAR scanner.

Mutual interference between LIDAR scanners is expected to become an increasingly important topic. It is due the fact

TABLE I: Occurrence of consecutive mutual interferences in a line

Arrangement	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Side by side #1	36 593	1467	409	150	74	43	25	12	1	0	0	0	0	1	0	1	0
Side by side #2	218 340	3181	810	344	151	108	73	51	1	2	1	0	1	0	0	1	1
Face to face	28 042	1908	522	222	123	58	37	35	3	0	1	0	0	0	1	0	0
Back to back a	4416	136	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Back to back b	4500	152	16	4	1	0	0	0	0	0	0	0	0	0	0	0	0



(a) consecutive interferences in a line

(b) Consecutive interferences at the same angle

Fig. 8: Occurrence of consecutive interferences

that, on the one hand, LIDAR scanners are used increasingly for obstacle detection system for autonomous vehicles. On the contrary, traffic density and percentage of vehicles equipped with LIDAR scanners increase considerably.

It is necessary to study how the mutual interference occurs depending on the reflectivity and material of objects. It is also a need to examine how the mutual interference occurs when quite a number of LIDAR scanner operate concurrently in a real traffic road environments. Mutual interference mechanisms have to be precisely understood in order to design and to verify the effectiveness of countermeasures to minimize mutual interference.

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